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Management of Innovation and Technology

Autodesk Fusion 360

Project Report

Project 1 – sand-casting (foundry) design

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1. Introduction

This report describes the complete design of the sand-casting cycle required to produce a steel housing component, starting from the finished part geometry assigned for the project. The work follows the standard foundry design methodology: beginning from the machined ("finished") part, the casting blank is defined by reintroducing the technological features required by the casting process (draft angles, fillet radii, machining allowances, cores), after which the pattern, feeding system, gating system, molding scheme and metalostatic verification are developed in sequence.

The component is a compact angled junction housing of the elbow/manifold type. It comprises two rounded-square bolting flanges, each carrying four corner bolt holes and a central circular machined bore, oriented at an angle (approximately 90°) to one another so that the two ports face in different directions. The two bores communicate internally through a central cored body. A third connection is provided at the bottom by a cylindrical spigot (with a small radial hole), while the central body is reinforced by rounded lateral bosses that blend into the flange regions. The part therefore routes and connects flow between three ports, two flanged and one cylindrical, and is the type of body typically used as a pump, valve or piping junction housing.

From a casting standpoint this geometry is demanding: the two flanges lie on non-parallel planes, the internal passages are curved and intersecting, and the wall thickness varies appreciably between the slender flange necks and the bulkier central junction. These features directly drive the key decisions made in the following sections — the choice of parting line and the elimination of undercuts, the feasibility of internal cores versus subsequent drilling, the distribution of machining allowances, and the location of risers for sound directional solidification. The overall part remains relatively small and moderately thin-walled, which also conditions the pattern tooling, the gating dimensions and the molding scheme. The geometry provided for the project is shown in Figure 1.

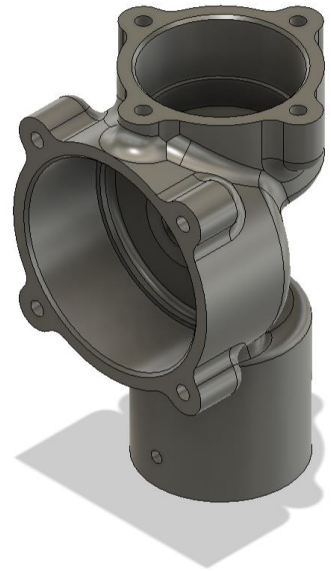


Figure 1. View of the component

2. Objectives

The objective of the project is to define every technological and economic element needed to manufacture the component by sand casting, delivering a coherent set of drawings and calculations. The work is organised into the following design activities, each producing a specific deliverable:

Activity	Deliverable	Main factors considered
Casting blank design	Casting-blank drawing	Parting line, cores vs. drilling, fillet radii, wall-thickness variation, machining allowances
Pattern design	Pattern drawing	Pattern type/material, shrinkage allowance, core prints
Feeding system (risers)	Number, size, position of risers	Directional solidification, feeding capacity, feeding radius
Gating system	Sprue, runner and gate cross-sections	Filling time, metal velocity, pressurized ratio
Molding and flasks	Molding method, flask type and size	Sand choice, multiple-cavity layout, flask dimensions
Metallostatic verification	Flask weighting / clamping check	Cavity geometry, cope weight, cores

3. Material Selection

The choice of material is governed by two criteria: the service requirements of the part (mechanical loads, operating environment and service conditions) and the machinability of the material in the subsequent chip-removal operations, since several surfaces will be finished by machining.

The component is specified as steel, and for a sand-cast steel part of medium mechanical strength the grade FeG 520 (cast steel, ≈ 520 MPa tensile strength) is adopted. This grade offers a good balance between mechanical performance and castability/machinability and is well suited to a housing subjected to moderate mechanical and pressure loads.

4. Component Data

The mass properties of the finished part, taken from the CAD model, are summarised below. These values are the reference baseline; the casting-blank volume and mass will be larger once draft angles and machining allowances are added.

Property	Value
Material	Steel (cast steel, FeG 520 assumed)

Overall envelope (L × W × H)	75.00 × 79.92 × 129.00 mm
Volume	152800 mm ³ ≈ 0.1528 dm ³
Surface area	54984 mm ²
Mass	1199 g ≈ 1.20 kg
Density	≈ 7.85 g/cm ³
Number of faces	140

4.1 Finished Part Drawing

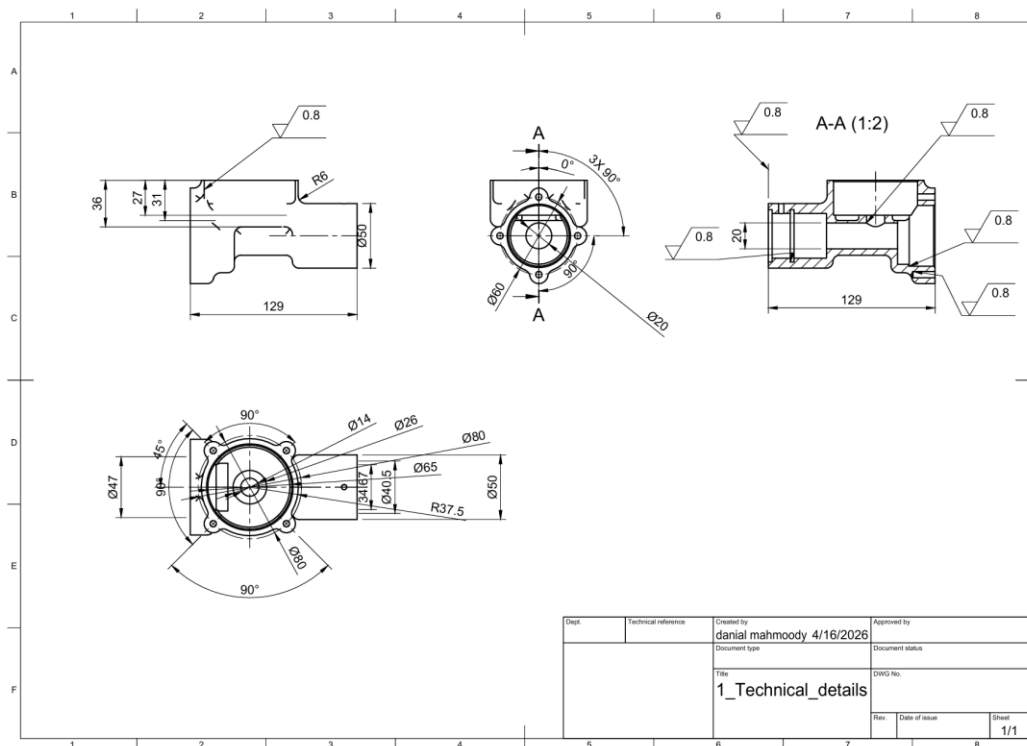


Figure 2. Finished part drawing, orthographic views and section A–A of the fully-machined component.

The finished (fully-machined) component is defined by the orthographic drawing of Figure 2, which provides the complete dimensional and surface-finish specification used as the starting point for the casting-blank design. The part is presented in three principal views together with a longitudinal section:

- a **side elevation** (upper left) giving the overall length of 129 mm and the stepped outer profile of the body, with the step heights 31, 27 and 36 mm and the corner blend R6;
- an **end view of the main flange** (centre), on which the cutting plane A–A is indicated and the bolt pattern is referenced — four holes at 90° spacing measured from the 0° axis, on the Ø60 bolt circle — together with the Ø20 radial cross-port on the spigot;

- the **longitudinal section A–A** (upper right, scale 1:2), which reveals the internal stepped bores and wall structure along the 129 mm length, with the seat depth of 20 mm and the concentric machined diameters;
- the **flange-face view** (lower left), showing the rounded-square mounting flange with its four corner bolt holes (Ø14 through, counterbored Ø26), the corner radius R37.5, the Ø80 flange outline, and the concentric machined bores Ø40.5 / Ø47 / Ø50 of the central port, located by the 34.67 mm dimension and the 45°/90° angular references.

The component is the angled three-port junction housing introduced in Section 1: a central body carrying two flanged ports oriented at approximately 90° to one another, plus a lower cylindrical spigot with a small radial hole. All functional surfaces — the flange faces, the bores and the seats — are specified to a surface roughness of Ra 0.8 µm (the 0.8 callouts distributed across the views) and are obtained by machining; the cast blank must therefore carry the corresponding stock on each of these surfaces, while the small holes and fine internal detail are deferred entirely to the machining cycle. The principal finished dimensions read from this drawing are summarised below.

Quantity	Value
Overall body length	129 mm
Flange outline (rounded square, across)	Ø80 mm
Flange corner radius	R37.5 mm
Bolt holes	4 × Ø14 through, counterbored Ø26, at 90° (Ø60 circle)
Central port bores	Ø40.5 / Ø47 / Ø50 mm
Spigot radial cross-hole	Ø20 mm
Outer step heights	31 / 27 / 36 mm
Seat depth	20 mm
General fillet	R6 mm
Surface finish (machined faces)	Ra 0.8 µm

4.2 Cutting-Plane and Undercut (Coring) Analysis

Before the casting blank can be defined, the finished part is sectioned by a cutting plane so that its interior can be examined and the features that obstruct moulding can be identified. The goal of this step is to locate

the **undercuts**, the re-entrant internal cavities and bores that lie behind the parting plane and would lock the pattern in the sand, and to decide how each is to be released: by a sand core, or by leaving it solid and producing it later by machining. The analysis is shown in two views in Figure 3.

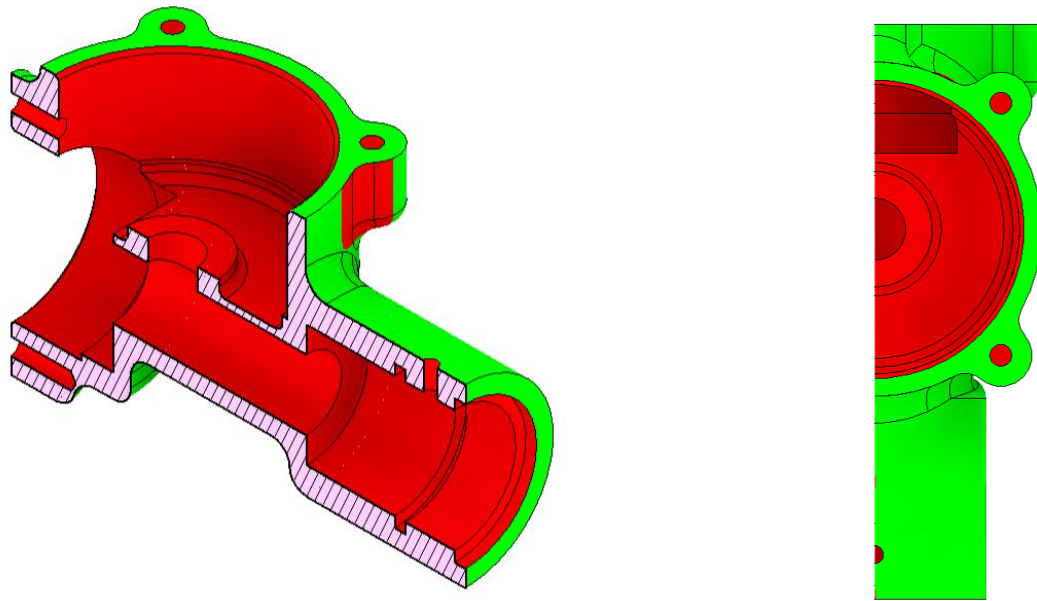


Figure 3. Cutting-plane analysis, sectioned body. Green: retained cast surfaces; red: internal cavities and undercuts to be removed; hatching: the cut through the metal.

In both views the colour coding has the same meaning: the green surfaces are the external cast faces that are kept on the blank, the hatched faces are where the cutting plane passes through the solid metal, and the red regions are the internal material that cannot be obtained as-cast by simple pattern withdrawal. Figure 3 (the left one) exposes the full internal passage system: the curved cavity that connects the upper flanged port, the lower spigot and the side port all meet at the central junction, and the intersecting curved bores there form clear undercuts. Figure 3 (the right one) looks onto the flange face and reveals the depth and the concentric steps of the central port, together with the two bolt holes; it confirms that the central cavity is deep and re-entrant relative to the flange parting plane.

From this examination each red region is classified. The small-diameter and long, slender passages (which would require fragile, unsupported cores) are filled solid and produced afterwards by drilling and boring, in line with the core-feasibility rules applied later in the blank design. The larger internal cavities that are both feasible and necessary for the part's function are instead formed by cores, which reproduce the through-ports and the branch cavity while allowing the pattern to be drawn cleanly from the mould.

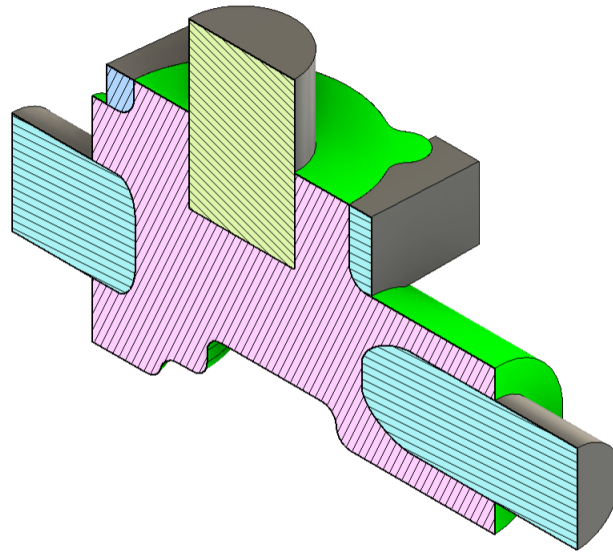


Figure 4. Resulting mouldable geometry after coring — the red material removed and the cores in place. Blue: side-port cores; yellow: upper-branch core; magenta: section through the cast metal; green: retained cast surfaces; grey: core prints.

Figure 4 shows the shape that is obtained once the cores have been produced and the highlighted (red) material removed. The hatched coloured bodies are the cores that form the internal passages: the two side cores (blue) produce the horizontal through-ports, and the upper core (yellow) produces the vertical branch cavity; the magenta hatching is the section through the cast metal itself, the green areas are the retained cast surfaces, and the grey ends are the core prints that seat and locate each core in the mould. This is the part that results after the cores, together with the filling of the unfeasible passages and the addition of machining stock, have been applied. It is on this cored geometry, free of locking undercuts, that all of the subsequent analyses are carried out: the wall-thickness (Heuver) check, the fillet radii, and the thermal-modulus, feeding, gating and metallostatic calculations that follow.

5. Casting Blank Design

5.1 Treatment of Small Holes (Drilling vs. Casting)



Figure 5. Finished part with the small holes identified, which are removed from the cast blank and obtained later by machining

Before developing the casting blank, the small-diameter holes present on the component were identified and suppressed from the cast geometry, as highlighted in Figure 5. These include the bolt holes on the two flanges and the small radial hole on the lower cylindrical spigot.

The reason for removing them is technological and economic. Producing such small-diameter holes directly in the foundry would require correspondingly small sand cores or core prints, which is both economically inefficient (extra core boxes, core making, setting and inspection for every cavity) and technically risky: thin cores of this size are fragile, are easily eroded or shifted by the metal stream during pouring, can generate gas-related defects, and are difficult to vent and to position accurately. The dimensional accuracy and surface finish required on these holes (especially the bolt holes, which must match a mating flange pattern) would in any case not be achievable as-cast.

For these reasons the small holes are not cast; they are produced during the subsequent machining cycle by drilling and reaming operations, which guarantee the required diameter, position tolerance and finish at low cost. Accordingly, in the casting blank these regions are left as full (solid) material, and the corresponding stock is removed later by machining. Only the large central bores, which are large enough to justify a robust core, are evaluated for casting.

5.2 Suppression of Machined Internal Features

In addition to the small holes, the finer internal machined features were also suppressed from the cast blank, as shown in Figure 6. These include the precision recesses and seats on the flange faces (the annular grooves/spotfaces visible around each bore), and the stepped internal diameters — the counterbores and shoulder steps inside the spigot and central body (highlighted in the section views).

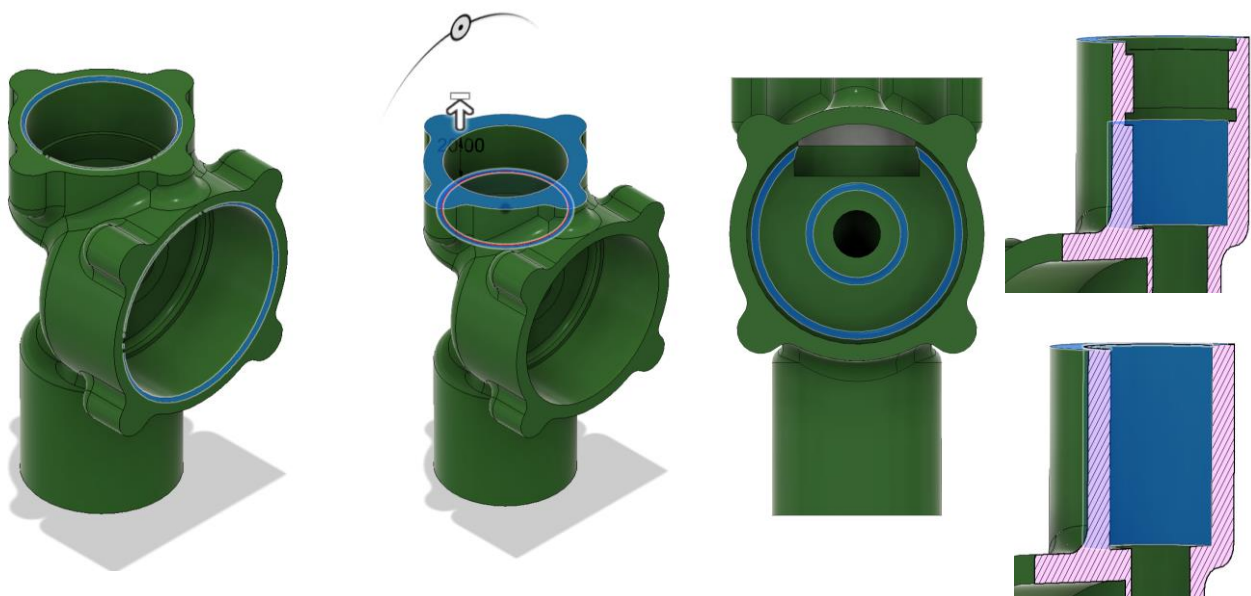


Figure 6. Internal machined features removed from the cast blank: flange-face seats/recesses and internal bore steps (counterbores), shown in plan and in cross-section

These features share the same justification as the small holes: their tight dimensional tolerances and surface-finish requirements cannot be achieved as-cast, and reproducing them in the mould would demand intricate, fragile core detail that is uneconomical and prone to defects. The sharp internal steps and shallow seats would also create local thin sections and abrupt thickness changes that promote shrinkage and hot-tearing problems during solidification.

Accordingly, these regions are left as simple, full (solid) volumes in the casting blank, carrying the appropriate machining allowance, and the final geometry (recesses, counterbores, shoulders and accurate bore diameters) is generated in the machining cycle by turning, boring and facing operations. This keeps the as-cast shape as smooth and continuous as possible, which both simplifies the mold and improves casting soundness, while deferring all high-precision detail to machining.

5.3 Machining Allowances on Internal Surfaces

The internal surfaces that will be finished by machining must be left with extra stock on the cast blank so that the final dimensions and surface finish can be obtained by chip removal. The required amount of stock is taken from Table 4 (Machining Allowances for Sand-Cast Unalloyed Steel Castings, UNI 6325-73), which gives the allowance as a function of the maximum casting dimension and the chosen accuracy grade.

Max casting size (mm)	≤ 80			80-180			180-315			315-500			500-800			800-1250			1250-1600			1600-2500		
	A	B	C	A	B	C	A	B	C	A	B	C	A	B	C	A	B	C	A	B	C	A	B	C
≤ 180	6	3	4	6	4	5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
180-500	6	4	5	7	5	5	8	6	6	10	7	7	-	-	-	-	-	-	-	-	-	-	-	-
500-1250	7	5	5	8	6	6	9	7	7	11	8	8	12	9	8	13	10	-	-	-	-	-	-	-
1250-2500	8	7	6	9	8	7	10	9	8	12	10	9	13	11	9	14	12	10	15	13	-	17	14	-

For this component the maximum casting dimension is approximately 129 mm, which places the part in the 80–180 mm size class of Table 4. Considering a medium-tolerance, repeated-production part, accuracy grade B is selected; for this size class and grade the recommended allowance is $s = 4$ mm, which is applied to the internal machined surfaces.

This 4 mm allowance was added uniformly to each internal feature. In the CAD model this was implemented efficiently by defining a single thickness parameter (a user variable in Fusion) equal to the 4 mm allowance, and referencing that variable for every internal surface. Using one driving variable keeps the allowance consistent across all features and allows it to be changed globally in a single step if the grade or size class is revised.

Figures 7a and 7b show the effect of the operation on a representative internal section: the original (finished) internal profile and the modified profile after the allowance has been added, with the added stock highlighted (hatched region) representing the material that will be removed during machining.

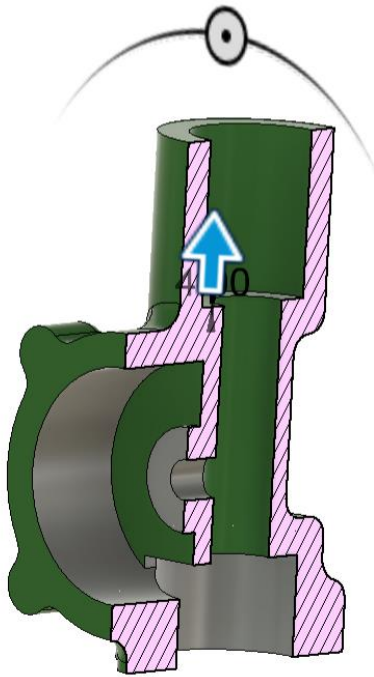


Figure 7a. Internal section before adding the machining allowance (finished internal profile)

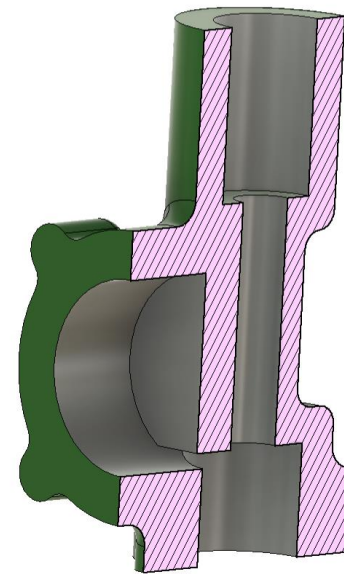


Figure 7b. Internal section after adding the 4 mm machining allowance; the hatched region is the added stock to be machined away

The result is a casting blank whose internal cavities are deliberately undersized relative to the finished bores, ensuring that sufficient, uniform material is available for the boring and facing operations while keeping the as-cast wall sections smooth and free of abrupt transitions.

5.4 Wall-Thickness Variation and Central-Core Feasibility

After the machining allowances were added to the internal surfaces, the wall-thickness distribution in the central region of the component was reviewed using the thickness-variation (Heuver-circle) criterion together with the core-feasibility rules of Table 2 (minimum diameter of as-cast holes versus hole length and type). The purpose of this step is to decide whether the central passages can be obtained directly in the foundry by means of a sand core, or whether they must be cast solid and produced later by machining.

Two issues emerged. First, once the allowance stock was added, the remaining cavities became too narrow relative to the surrounding wall, so that the local thickness variation no longer satisfied the smooth-transition requirement of the Heuver method (adjacent sections differing by more than $\approx 30\%$), which would promote uneven cooling and shrinkage stresses. Second, the resulting cavity proportions failed the castability conditions of Table 2, meaning the corresponding cores would be too slender and too long to survive pouring and solidification. For both reasons, the central region was filled solid in the casting blank, and the final internal geometry will be produced by drilling/boring in the machining cycle. The progressive result of this decision is the simplified, solid central section shown in Figure 8d.

Short upper passage (Figure 8a). Measured values: $D = 10$ mm, surrounding wall $s = 18$ mm, length $L = 12$ mm. Here $D < 2s$ ($10 < 36$), so the governing condition is $L \leq D = 10$ mm for a through-hole. The actual length $L = 12$ mm exceeds this limit, and the diameter is in any case very small; a core of this size would be fragile and unreliable. The passage is therefore filled solid and will be drilled later.

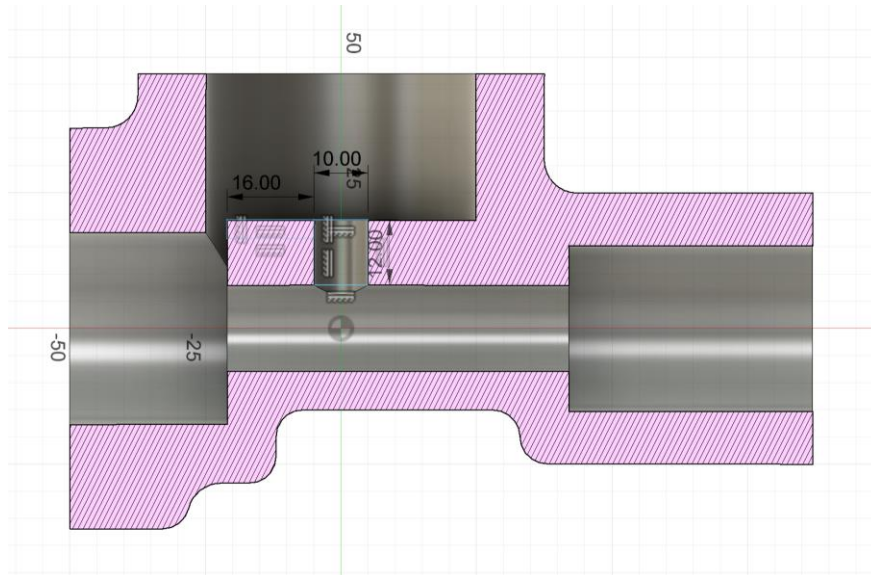


Figure 8a — Short passage, $D = 10$, $s = 18$, $L = 12$ mm: fails Table 2 ($L > D$), to be filled

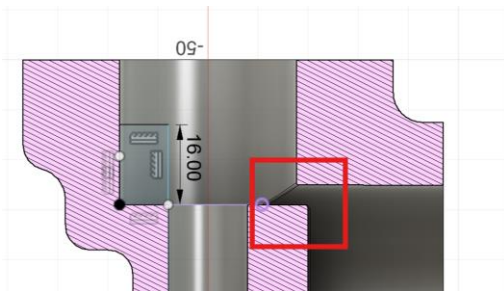


Figure 8b. Small local pocket (red box): too small/fragile to core, to be filled

Small local pocket (Figure 8b). A further small feature (highlighted by the red box) is too small and shallow to be reliably formed as part of the upper (cope) core: such a small projection would be mechanically weak, difficult to vent and position, and easily eroded or broken by the metal stream, so it does not meet the practical robustness required of a cored detail. In line with the same principle applied to the other small features, this pocket is filled solid and obtained later by machining.

Central through passage (Figure 8c). Measured values: $D = 16$ mm, wall thickness $s = 7$ mm, length $L = 79$ mm. Since $2s = 14$ mm and $3s = 21$ mm, the case falls in the band $2s \leq D \leq 3s$, for which a through-cast hole is admissible only if $L \leq 3D = 48$ mm. The actual length $L = 79$ mm greatly exceeds this limit, so a core in this passage would be excessively long and unsupported. The passage is therefore filled solid and will be machined afterwards.

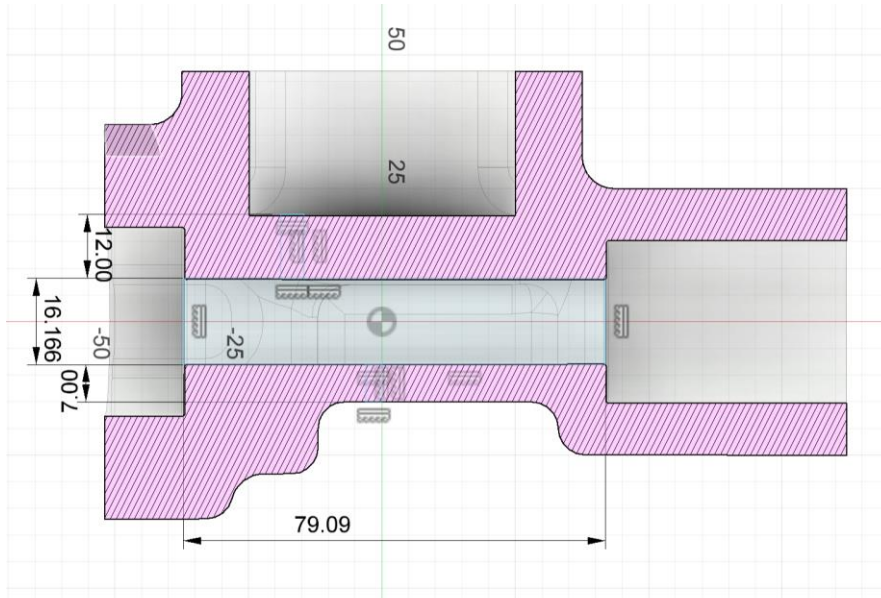


Figure 8c. Central through passage, $D = 16$, $s = 7$, $L = 79$ mm: fails Table 2 ($L > 3D$), to be filled

Resulting blank section (Figure 8d). With all three regions filled, the central body of the casting blank becomes a continuous solid mass, free of the slender internal cavities that could not be reliably cast. This solid section is easier to mould, improves casting soundness and feeding, and defers all internal precision geometry to the machining operations.

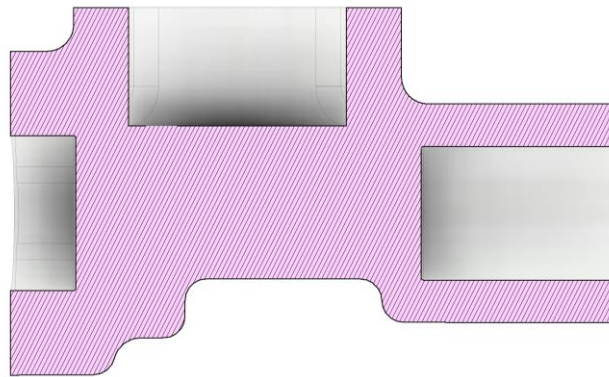


Figure 8d. Final central section of the casting blank after filling (solid body)

5.5 Verification of Wall-Thickness Variation (Heuver-Circle Method)

Having defined the filled central body, the wall-thickness distribution of the casting blank was verified using the Heuver inscribed-circle method. Circles are inscribed tangent to the walls at successive sections along the part; the criterion requires that the diameter of tangent circles in adjacent sections vary by no more than approximately 30 %, so that neighbouring regions cool at comparable rates and shrinkage stresses and hot-tearing at abrupt thickness changes are avoided.

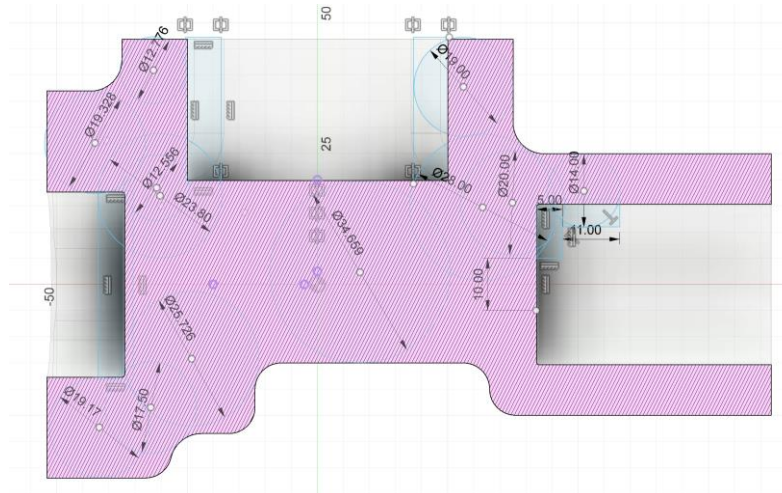


Figure 9a. Heuver inscribed circles applied to the casting-blank section to check thickness variation

Checking the diameters between adjacent stations, several transitions exceeded the 30 % limit — in particular the growth toward the bulky central junction. These abrupt increases indicate hot-spots at the junction and unfavourable cooling gradients, which would not satisfy the smooth-progression requirement.

To bring the variation within the allowable range, the cores defining the adjacent cavities were reduced/reshaped, locally thinning the over-thick sections and smoothing the transition between neighbouring regions so that the inscribed-circle diameters change more gradually from section to section. The reshaped blank, obtained after enforcing the thickness-variation rule, is shown in Figure 9b. In this final geometry the section progresses more uniformly from the thinner outer regions toward the central body, which improves directional cooling and reduces the risk of shrinkage defects, while remaining compatible with the previously established core-feasibility decisions.

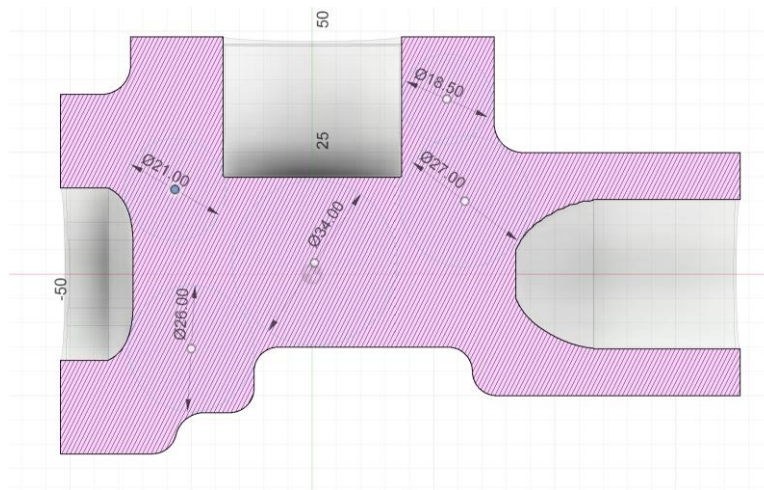
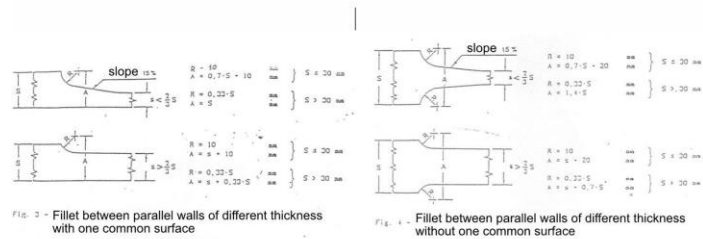


Figure 9b. Casting-blank section after reshaping the cores to satisfy the thickness-variation criterion

5.6 Fillet Radii

All sharp internal corners and junctions of the casting blank were rounded with fillets. In casting, fillets serve three purposes: they reduce stress concentration during cooling, limit local metal accumulation

(avoiding hot-spots that solidify last and lead to shrinkage porosity), and prevent mould-erosion zones where a sharp sand corner would otherwise be washed away by the metal stream. The general reference for sizing fillet radii as a function of the local wall thickness is Table 3.



Case	R	Case	R
	1,25 s		s
	s		10
	1,5 s		0,25 s

For this project a constant fillet radius of $R = 0.5\text{ mm}$ was adopted at the internal junctions, as specified for the exercise. The fillets were applied in the CAD model using the constant-radius fillet feature with a tangent-chain selection and a rolling-ball corner type (Figure 10a), so that the rounding propagates smoothly along connected edges and blends tangentially into the adjacent faces. The resulting casting-blank section, with the internal junctions rounded, is shown in Figure 10b.

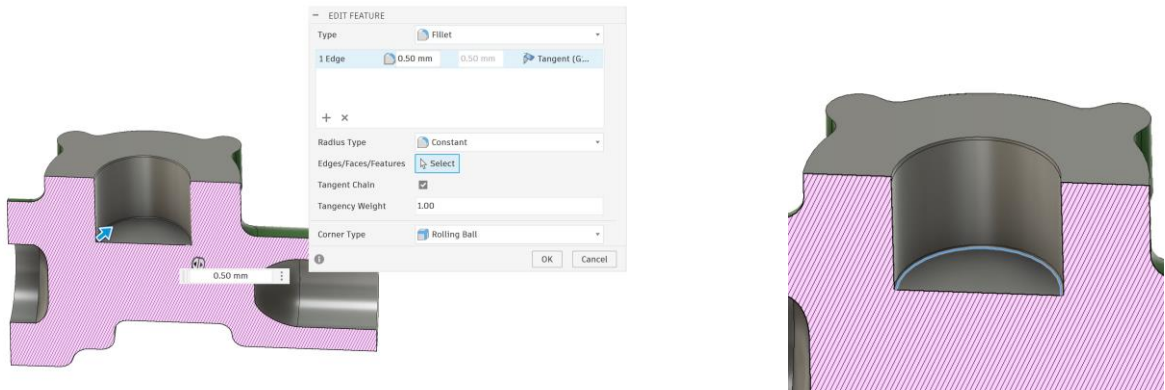


Figure 10a. Application of the constant-radius fillet ($R = 0.5\text{ mm}$) in CAD, with tangent chain and rolling-ball corner type

Figure 10b. Casting-blank section after applying the fillet radii at the internal junctions

The rounded transitions remove sharp re-entrant corners from the as-cast geometry, giving the mold smoother, more erosion-resistant surfaces and reducing local stress concentration at the junctions, while remaining consistent with the thickness-variation criterion established in the previous step.

5.7 Draft Angles

The parting line is chosen on the flat plane that splits the casting into a cope and a drag while eliminating all undercuts — the re-entrant features identified in the sectioning analysis that would otherwise prevent the pattern from being withdrawn from the sand.

Draft angles are then added to the blank. A draft angle is a slight taper given to the walls that lie along the pattern-withdrawal direction, so that the pattern (and the external cores) can be drawn from the compacted

sand without dragging or tearing the mould. The draft is therefore applied to the external core surfaces and to the other external surfaces of the component that run parallel to the pull direction.

The amount of draft is taken from Table 1 (recommended draft angle as a function of wall height). The drafted walls of this component fall in the 30–50 mm wall-height band, and the free-pattern column of that band gives a minimum draft of 3° . Accordingly, a single draft angle of:

$$\gamma = 3^\circ$$

is adopted and applied uniformly to all drafted surfaces of the part and its cores. This is consistent with the casting-blank drawing, on which the tapered walls are dimensioned at 93° (i.e. $90^\circ + 3^\circ$).

Because the draft is added on top of the nominal geometry, the inclined surfaces it creates on the functional faces lie within the machining allowance and are removed in the subsequent chip-removal cycle; the draft therefore eases pattern and core extraction without altering any finished dimension.

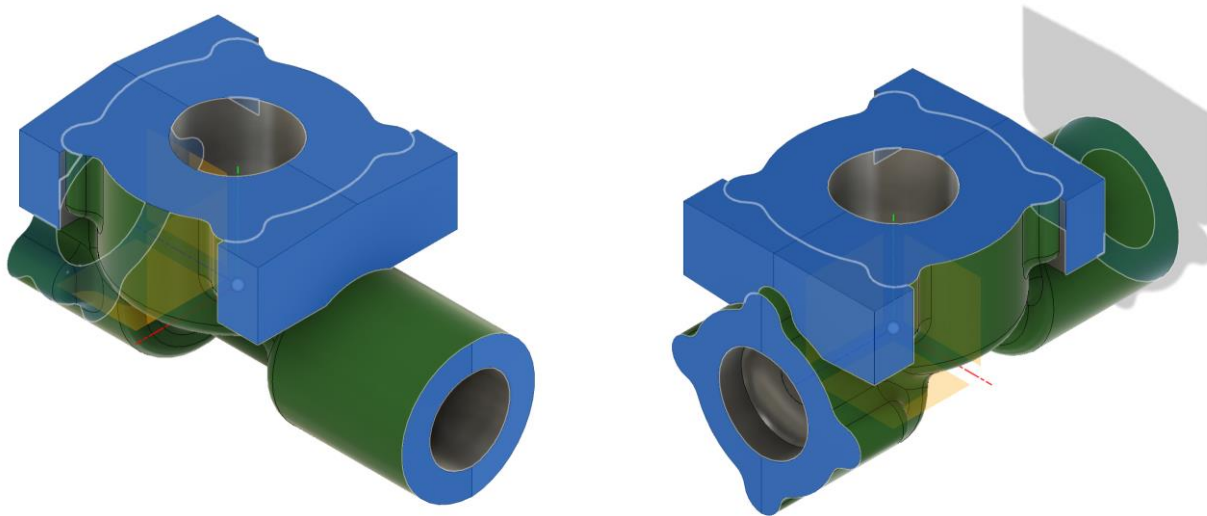


Figure 11. Component with the 3° draft applied, shown across the parting plane — tapered external walls and core surfaces, with the central cored bore.

Figure 11 shows the component after the 3° draft has been applied, viewed across the parting plane (the translucent plane). The cope-side and drag-side volumes are distinguished by colour, and the tapered walls of both the body and the cores are visible, together with the central cored bore. With the undercuts eliminated by the parting line and the walls drafted, the pattern can be withdrawn cleanly and the cores set from above, completing the mouldable definition of the casting blank.

5.8 Casting Blank Drawing

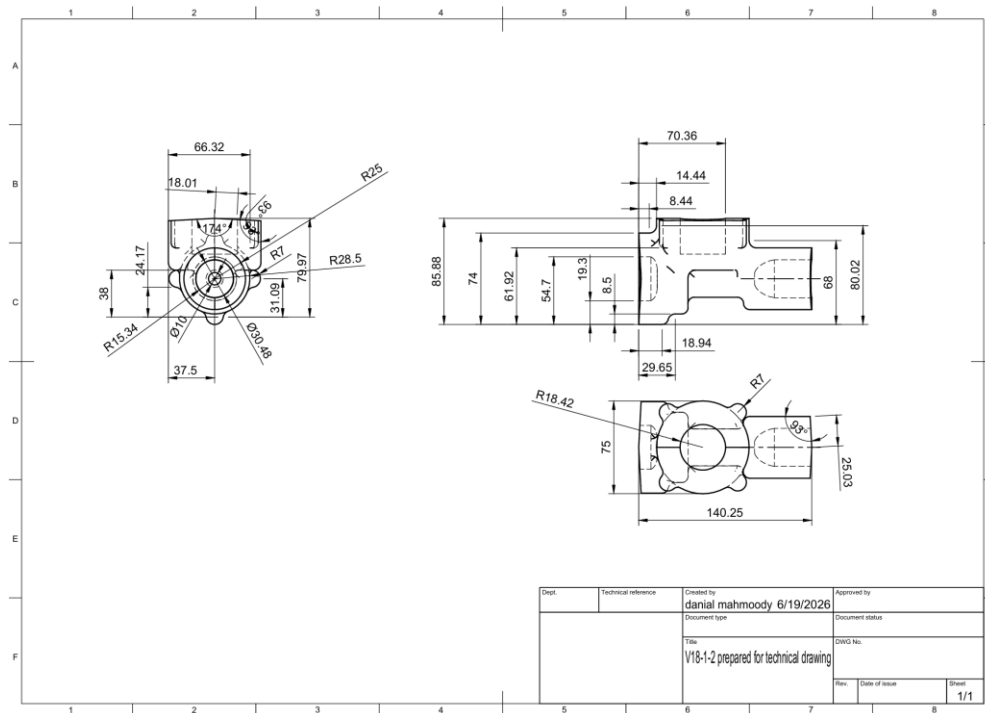


Figure 12. Casting-blank drawing, the component prepared for casting, with draft angles, casting fillets and machining allowance applied.

Figure 12 shows the casting blank obtained at the end of the blank-design activity: the finished geometry of Figure 12 after the precision holes and internal machined features have been suppressed, the machining allowances added, the unfeasible central passages filled solid, the junctions filleted and the draft angles applied for pattern extraction. This is the geometry that is actually moulded and poured, and it is the input to the thermal analysis that follows. Compared with the finished part, the blank differs in the following respects, all visible in the drawing:

- **Overall size has grown:** the body length is now 140.25 mm (against 129 mm finished) and the flange envelope measures about 75×85.88 mm, the increase being the machining stock added to the surfaces that are finished later;
- **Draft angles** have been added to every wall lying along the withdrawal direction, shown by the tapered profiles and the 93° and 174° angular callouts (a nominal $\approx 3^\circ$ draft per face), so that the pattern can be drawn cleanly from the sand;
- **Generous casting fillets and corner radii** replace the sharp machined edges — R25, R28.5, R18.42, R15.34 and R7 blend the arms into the central body and round the flange corners — which limits stress concentration, avoids isolated hot-spots and protects the mould from erosion;
- **The small bolt holes and fine internal features have been removed** (left full), leaving only the large central cavity as a cast/cored feature (the $\text{Ø}10 / \text{Ø}30.48$ central detail); all precision diameters and seats are deferred to machining, giving smooth, continuous as-cast surfaces;
- **The mounting flanges are rendered as rounded squares** (corner radii R18.42 / R7), again easing moulding and improving metal flow into the cavity.

The three views — the flange end view (upper left), the side elevation of the branched body (upper right) and the plan of the rounded-square flange and spigot (lower right), together define the blank that is subdivided into thermal zones in the next section. The resulting solid, smoothly-blended body is deliberately simpler than the finished part: it is easier to mould, feeds more soundly, and confines all high-precision geometry to the subsequent machining cycle. The principal blank dimensions are summarised below.

Quantity	Value
Overall body length	140.25 mm
Flange width	75 mm
Overall height (elevation)	85.88 mm
Draft callouts ($\approx 3^\circ$ per face)	$93^\circ / 174^\circ$
Casting fillets / blend radii	R7 / R15.34 / R18.42 / R25 / R28.5
Flange corner radii	R18.42 / R7
Retained central cast feature	$\varnothing 10 / \varnothing 30.48$ mm
Branch / arm reference dims	70.36, 68, 80.02, 61.92, 54.7 mm

5.9 Thermal Analysis

To prevent shrinkage cavities in the casting, the riser must be the last portion of the assembly to solidify. The onset of solidification of each region is proportional to its thermal modulus, defined as the ratio of the volume to the heat-transfer surface area:

$$M = \text{Volume} / \text{Heat-transfer surface area}$$

The part was subdivided into four zones (A–D). For each zone the volume and the surface area were obtained from the body Properties panel in Autodesk Fusion.

Zone layout

The component is a multi-branch body in which three arms converge on a central hub:

- Zone A — right-hand horizontal arm with a through-bore;
- Zone B — central hub / junction where the three arms meet;
- Zone C — upper vertical branch with a bored pocket (enlarged in the re-correction, see below);
- Zone D — left-hand horizontal arm with a blind pocket.

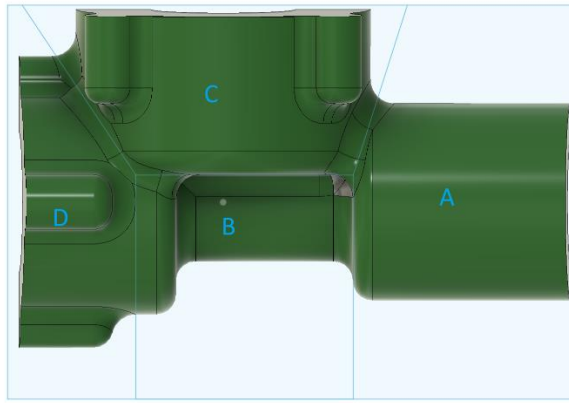


Figure 13. Zone subdivision (A, B = hub, C, D).

5.9.1 Cooling surface (heat-transfer area)

The area reported by Fusion for each isolated zone is the *total closed-body area*, which also includes the flat faces shared with the adjacent zones. These interface faces lie metal-against-metal at almost the same temperature, so they exchange no heat with the mould and must be subtracted. Only the faces in contact with the sand are retained as the heat-transfer surface S . The interface areas removed for each zone are listed below.

Zone calculations

Zone A

- $V_A = 75,552.9 \text{ mm}^3$
- $S_A = 15,739.704 - (967 + 699) = 15,739.704 - 1,666 = 14,073.7 \text{ mm}^2$

$$M_A = 75,552.9 / 14,073.7 = \mathbf{5.4 \text{ mm}}$$

Zone B

- $V_B = 42,831.4 \text{ mm}^3$
- $S_B = 10,089.821 - (967 + 2,413 + 1,660) = 10,089.821 - 5,040 = 5,049.8 \text{ mm}^2$

$$M_B = 42,831.4 / 5,049.8 = \mathbf{8.5 \text{ mm}}$$

Zone C

- $V_C = 155,900 \text{ mm}^3$
- $S_C = 22,721.599 - (2,413 + 1,323 + 699) = 22,721.599 - 4,435 = 18,286.6 \text{ mm}^2$

$$M_C = 155,900 / 18,286.6 = \mathbf{8.5 \text{ mm}}$$

Zone D

- $V_D = 64,595.2 \text{ mm}^3$
- $S_D = 12,473.401 - (1,660 + 1,323) = 12,473.401 - 2,983 = 9,490.4 \text{ mm}^2$

$$M_D = 64,595.2 / 9,490.4 = \mathbf{6.8 \text{ mm}}$$

5.9.2 Summary of thermal moduli

Zone	Volume V (mm ³)	Total area (mm ²)	Cooling surface S (mm ²)	M = V/S (mm)
A	75,552.9	15,739.7	14,073.7	5.4
B	42,831.4	10,089.8	5,049.8	8.5
C	155,900	22,721.6	18,286.6	8.5
D	64,595.2	12,473.4	9,490.4	6.8

5.9.3 Directional solidification and riser location

For sound directional solidification the casting must freeze progressively toward the riser, which has to be the last element to solidify. Two conditions are applied to the thermal moduli of adjacent zones along the path that leads to the riser:

- The modulus must increase by at least ~10% from one zone to the next toward the last-solidifying region, to guarantee a directional front;
- The difference between adjacent moduli should not exceed ~30%, to avoid the steep thermal gradients that cause shrinkage stresses and hot tears.

The riser is placed on top of branch C, the natural feed channel into the central hub. Checking the moduli along the feed path (arms → hub B → neck C → riser):

$$M_A (5.4) < M_D (6.8) < M_B (8.5) \approx M_C (8.5)$$

- D → B: $6.8 \rightarrow 8.5 = +25\%$ (increasing and within the 10–30% band ✓)
- A → B: $5.4 \rightarrow 8.5 = +58\%$ (increasing (directional) but above the 30% guideline; a thin arm meets the heavy hub directly)
- B → C: $8.5 \rightarrow 8.5 \approx +1\%$ (the hub and the feed neck are effectively balanced)

So, the path is directional overall (each step increases toward the riser), but two steps sit outside the ideal band: A→B is too steep, and B→C is essentially flat rather than the +10% margin wanted at the feed neck.

What the method prescribes when the conditions are not met. The thermal modulus is governed by the local volume-to-cooling-surface ratio, so where the geometry cannot be altered the solidification order is corrected by acting on the *cooling rate* of individual zones, using mould features that do not change the casting itself:

- Chills: metal inserts (steel, cast iron, graphite) placed in the sand against a zone to extract heat locally and accelerate its solidification, lowering its effective modulus. Used on a zone that is "too hot for its position" in the sequence.
- Insulating / exothermic sleeves: collars placed around a zone or the riser to slow cooling and delay solidification, raising the effective modulus of whatever must stay liquid longest.

Applied to the two non-ideal steps here:

- B → C balanced (+1%). A chill on the hub B tips it to solidify just before the neck, and an insulating/exothermic sleeve around the C branch and the riser keeps the feed channel liquid last. Together they convert the flat step into a directional one (the hub freezes, then the neck, then the riser) without altering the part. (Insulating the riser alone would not help, since the neck C below it would freeze first and choke the feed.)
- A → B steep (+58%). A jump above 30% from a thin arm into a heavy junction is mitigated by generous fillet radii at the junction — which also reduce stress concentration and metal accumulation at the reentrant corner — together with an end chill on arm A to sharpen the inward gradient.

End chills on the extremities (A and D) also give the casting a defined cold end, anchoring the start of the directional front and keeping the design on the safe side.

5.10 Riser Sizing

Single riser located on top of branch C, fitted with an insulating/exothermic sleeve. The riser must be the last element to solidify and must feed the controlling section of the casting.

5.10.1. Required riser modulus

The riser is sized off the controlling section it feeds ($M_B \approx M_C = 8.5$ mm), applying the sleeve effect and the safety factor:

- sleeve effect: $8.5 \times 1.1 \approx 9.4$ mm
- riser requirement: $M_{\text{riser}} \geq 1.2 \times 9.4 \approx 11.2$ mm

A direct modulus-based pick would force the first standard row with $M \geq 11.2$ mm, i.e. $M_M = 1.2$ cm (D = 64 mm, H = 96 mm, V = 315 cm³, W = 2.14 kg). For this part that riser is disproportionately large — its volume (315 cm³) is almost as big as the whole casting (339 cm³), and its 4% feeding capacity (0.79 L) is more than double what the part actually needs, which would wreck the casting yield. The riser is therefore sized on the casting's real shrinkage demand, with the sleeve relied on to supply the extra solidification time the smaller geometric modulus alone would not give.

5.10.2. Riser selection by feeding capacity — Table 6 ASSOFOOND (H = 1.5 D)

The selection is made by comparing the casting volume with the maximum-casting volume listed in the 4% shrinkage column of Table 6 (0.2% C steel, see step 3). The casting volume is:

$$V_C = 3.389 \times 10^5 \text{ mm}^3 = 338.9 \text{ cm}^3 = 0.339 \text{ L}$$

This value falls between two consecutive rows of the 4% column:

- $M_M = 0.9$ cm → max casting 0.330 L (too small)

- $M_M = 1.0 \text{ cm} \rightarrow \text{max casting } 0.450 \text{ L (sufficient)}$

Following the rule “*when the casting falls between two rows, take the larger*”, the $M_M = 1.0 \text{ cm}$ row is selected:

Quantity	Value
Modulus M_m	1.0 cm (10 mm)
Diameter D_m	54 mm
Height $H_m (= 1.5 D)$	81 mm
Volume V_m	180 cm ³ (0.180 L)
Weight W_m	1.22 kg

Compared with the modulus-based $M = 1.2$ riser, this cuts the riser metal from 315 cm³ to 180 cm³ ($\approx 43\%$ less), markedly improving yield while still covering the shrinkage of the part.

5.10.3. Feeding-capacity check

For 0.2% C steel poured at 1560 °C, Table 7 gives a solidification shrinkage of $\approx 4\%$. From the 4% column of the selected $M = 1.0$ row, this riser can feed a casting up to $V = 0.450 \text{ L} / W = 3.52 \text{ kg}$. The casting is $V_C = 0.339 \text{ L}$, i.e. $\approx 2.64 \text{ kg}$ in steel ($\gamma = 7.8 \text{ kg/dm}^3$):

Check	Casting	Riser limit (4%)
Volume	0.339 L	0.450 L
Weight	2.64 kg	3.52 kg

The casting uses about three-quarters of the riser’s rated feeding capacity ($\approx 1.3\times$ margin) — comfortably inside the limit while avoiding the heavily oversized $M = 1.2$ riser. The design is further on the safe side because of the end chills on Zones A and D, which steepen the thermal gradient toward the riser.

5.10.4. Riser neck — Table 8 (steel)

With $D = D_m = 54 \text{ mm}$, $d = 0.40D$ and $L = 0.16D$:

$$d = 0.40 \times 54 \approx 22 \text{ mm} \quad L = 0.16 \times 54 \approx 9 \text{ mm}$$

The 45° neck collar connects the riser to branch C, small enough to break off cleanly after casting, yet large enough to stay open until the riser has finished feeding.

5.10.5. Summary

A single cylindrical riser, $D = 54 \text{ mm} \times H = 81 \text{ mm}$ ($M_M = 1.0 \text{ cm}$), fitted with an insulating/exothermic sleeve, is placed on top of branch C and connected through a neck of $d = 22 \text{ mm}$, $L = 9 \text{ mm}$. Rather than being sized by the bare modulus rule (which would have forced an oversized $M = 1.2$ riser), it is selected from the casting's own volume against the 4% feeding-capacity column of Table 6: 0.339 L sits between the 0.9 cm and 1.0 cm rows, so the larger (1.0 cm) is chosen. The riser feeds the whole casting (0.339 L / 2.64 kg) inside its 0.450 L / 3.52 kg capacity, with the sleeve ensuring it remains the last section to solidify.

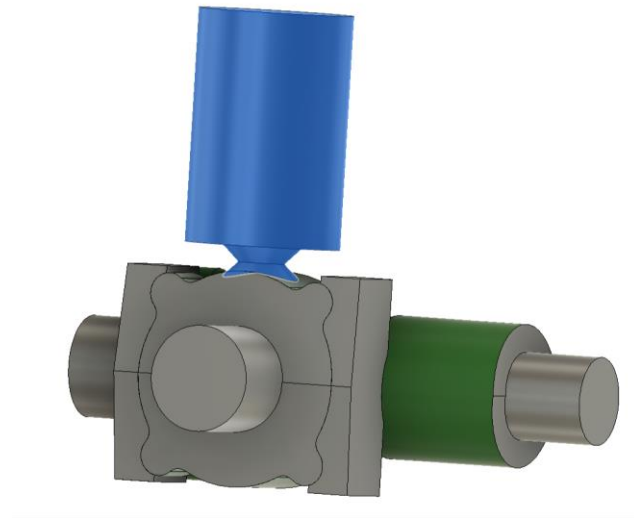


Figure 14. Riser

5.11 Gating System Sizing

Single non-pressurised-to-pressurised choke arrangement feeding the casting through one ingate on branch C. The system is sized from the poured weight and a target pouring time: the pouring time fixes the volumetric flow rate, the effective metallostatic head fixes the gate velocity, and the two together fix the choke (gate) area. Sprue and runner are then scaled from the gate through the pressurised gating ratio.

5.11.1. Poured weight and pouring time

The poured metal is the casting plus the riser. Using the as-modelled (Fusion) volumes:

$$V = V_C + V_{\text{riser}} = 338\,900 + 188\,900 = 527\,800 \text{ mm}^3 = 0.5278 \text{ dm}^3$$

$$W = V \cdot \gamma = 0.5278 \times 7.8 = 4.117 \text{ kg}$$

The pouring time follows the empirical square-root rule $t = 2\sqrt{W}$ (W in kg, t in s):

$$t = 2\sqrt{4.117} = 4.06 \text{ s}$$

The mean volumetric flow rate the system must deliver is therefore:

$$Q_m = V / t = 0.5278 / 4.06 = 0.1301 \text{ dm}^3/\text{s} \quad (1.301 \times 10^{-4} \text{ m}^3/\text{s})$$

Quantity	Symbol	Value
Poured volume (casting + riser)	V	0.5278 dm ³
Steel density	γ	7.8 kg/dm ³
Poured weight	W	4.117 kg
Pouring time ($t = 2\sqrt{W}$)	t	4.06 s
Mean flow rate	Q _m	0.1301 dm ³ /s

5.11.2. Effective head and gate velocity

The metal head falls as the cavity fills, so an average effective head is used. With the initial and final head heights measured in Fusion ($h_i = 125$ mm, $h_f = 118.7$ mm), the time-averaged head is obtained from the mean of $\sqrt{h}$:

$$H = (\sqrt{h_i} + \sqrt{h_f})^2 / 4 = (\sqrt{125} + \sqrt{118.7})^2 / 4 = 121.8 \text{ mm (0.1218 m)}$$

The gate velocity follows Torricelli's law with a discharge coefficient $C_d = 0.65$ to account for friction and turbulence losses:

$$V_G = C_d \cdot \sqrt{(2gH)} = 0.65 \times \sqrt{(2 \times 9.81 \times 0.1218)} = 1.005 \text{ m/s}$$

5.11.3. Choke (gate) area

In a pressurised system the choke is the ingate. Its required total area is the flow rate divided by the gate velocity:

$$S_G = Q_m / V_G = 1.301 \times 10^{-4} / 1.005 = 1.294 \times 10^{-4} \text{ m}^2 = 129.4 \text{ mm}^2$$

5.11.4. Gating ratio and channel areas

A pressurised ratio is adopted so that the gate remains the smallest section (the choke), keeping the system back-filled and the flow calm to limit reoxidation and dross:

$$S_S : S_R : S_G = 4 : 3 : 2$$

Taking the gate as the 2-part reference ($S_G = 129.4 \text{ mm}^2$, so 1 part = 64.7 mm^2):

Channel	Symbol	Parts	Required area
Gate (ingate)	SG	2	129.4 mm ²
Runner	SR	3	194.1 mm ²

Channel	Symbol	Parts	Required area
Sprue	SS	4	258.9 mm ²

5.11.5. Standard-section selection — Table 12

Each required area is rounded up to the nearest standard section in Table 12 (ASSOFOND). The runner and ingate sections are trapezoidal/triangular, so their tabulated cross-section is about half of the bounding A×B; the sprue is circular.

Part	Required	Selected section	Std. area	Result
Single ingate (gate)	129.4 mm ²	18 × 18 mm	162 mm ²	✓
Runner	194.1 mm ²	20 × 24 mm	240 mm ²	✓
Sprue	258.9 mm ²	Ø 20 mm	314 mm ²	✓

Every standard section sits at or above its requirement, so the as-built ratio is slightly over-sized on the sprue and runner relative to the gate — which keeps the gate firmly as the choke and preserves the pressurised character of the system.

5.11.6. Summary

A single-ingate pressurised gating system (ratio $S_S:S_R:S_G = 4:3:2$) feeds the casting on branch C. Sized for a poured weight of 4.117 kg and a 4.06 s pouring time, it requires a gate choke of 129.4 mm² at a gate velocity of 1.005 m/s under an effective head of 121.8 mm. The standard sections selected from Table 12 are:

- Sprue — Ø 20 mm (314 mm²)
- Runner — 20 × 24 mm trapezoidal (240 mm²)
- Ingate — 18 × 18 mm trapezoidal (162 mm²), single, on branch C

The selected sections all exceed their calculated minima, so the system delivers the design flow within the target pouring time while keeping the ingate as the controlling choke.

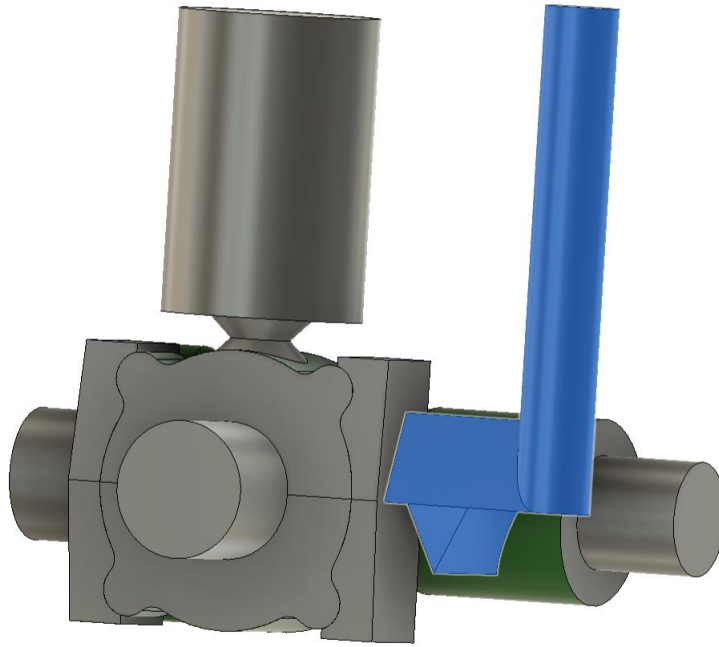


Figure 15. Gating System

5.12 Molding and Flask

Given the small size of the component and the production batch of 500 pieces, the mould is produced by pattern-plate (match-plate) molding using a synthetic, resin-bonded sand, which cures at room temperature and gives good surface finish and dimensional accuracy. A single casting is moulded per flask, gated through the single ingate on branch C and fed by the single riser placed on top, so that the sprue, runner, ingate and riser all lie on one flat parting plane.

The casting is located in the mould with the parting plane horizontal (cope above, drag below), using the flat parting surface selected earlier. The riser and the sprue/pouring patterns are mounted vertically on the cope side, giving top feeding and a clean pour. Figures 16 show, in sequence, the casting positioned in the flask cavity, the mould layout with the sprue and riser in place, and the closed flask ready for pouring.

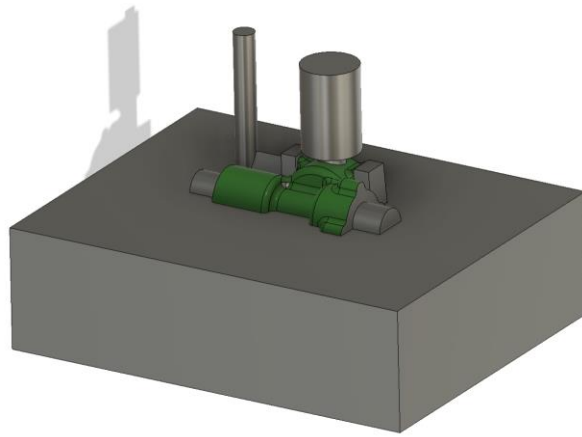


Figure 16a. The casting (green) located inside the flask cavity, surrounded by the moulding sand.

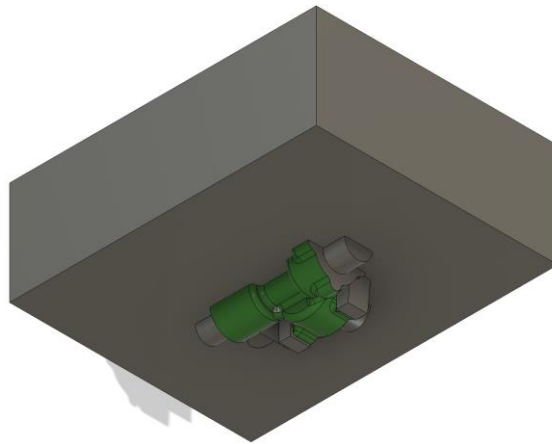


Figure 16b. Mould layout on the parting surface. the casting with the sprue pattern (thin) and the riser pattern (thick) standing vertically on the cope side.

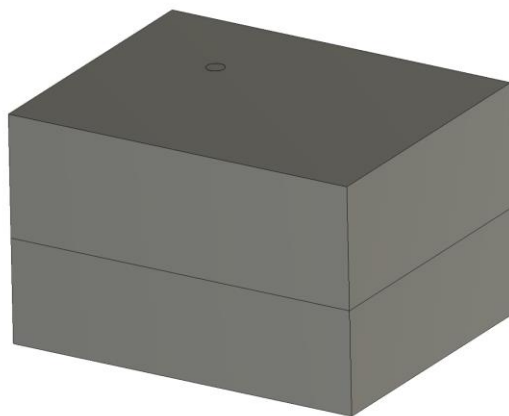


Figure 16c. The closed flask (cope + drag assembled) with the pouring opening on the top face, ready for pouring.

The flask size is selected from Table 11 (standard flask dimensions, UNI 6765-70). A rectangular flask of 315×400 mm in plan and 250 mm total height is adopted — large enough to contain the casting, the riser and the gating system with adequate sand thickness around the cavity. The upper (cope) portion used for the metallostatic check that follows is the top 135 mm of this flask.

Quantity	Value
Moulding method	Pattern-plate (match-plate)
Sand	Synthetic, resin-bonded (self-hardening)
Castings per mould	1 (single cavity)
Sprue / riser	1 each, mounted on the cope
Flask plan size (a × b)	315×400 mm
Flask total height	250 mm
Cope height (used in metallostatic check)	135 mm
Standard	UNI 6765-70 (Table 11)

5.13 Metallostatic Forces

During pouring the liquid metal exerts an upward (lifting) pressure on the cope. This metallostatic force must be counteracted by the weight of the upper flask; if it were not, weights would have to be added to the cope or the clamping screws sized accordingly. The check therefore compares the lifting force F_{TOT} against the sand weight of the cope F_S .

5.13.1. Method

The lifting force equals the weight of the metal column standing above the upward-facing surfaces of the casting in the cope:

$$F_{TOT} = \gamma \cdot \Sigma V_i \quad \text{with} \quad V_i = h_i \cdot r_i \cdot A$$

Following the simplified approach indicated by the course, each upward-facing surface is approximated by a rectangular projection rather than by its exact revolved (annular) area: r_i is the radial width of the surface, h_i is the height of metal above it up to the top of the flask, and $A = 315$ mm is the flask length. In effect the part's side profile is treated as extruded across the flask length instead of revolved about its axis. The steel specific weight is $\gamma = 7.8$ kg/dm³.

5.13.2. Lifting force on the cope

Applying the formula to each surface of the casting ($A = 315$ mm throughout):

Surface	h (mm)	r (mm)	A (mm)	$V = h \cdot r \cdot A$ (mm ³)
1	95.3	24.0	315	720 468
2	5.2	57.0	315	93 366
3	95.3	19.6	315	588 382
4	108.5	42.0	315	1 435 455
Total				2 837 671

$$F_{\text{TOT}} = \gamma \cdot V = 7.8 \times 2.838 = 22.1 \text{ kg}$$

(The volume $2\,837\,671 \text{ mm}^3 = 2.838 \text{ dm}^3$, using $1 \text{ dm}^3 = 10^6 \text{ mm}^3$.)

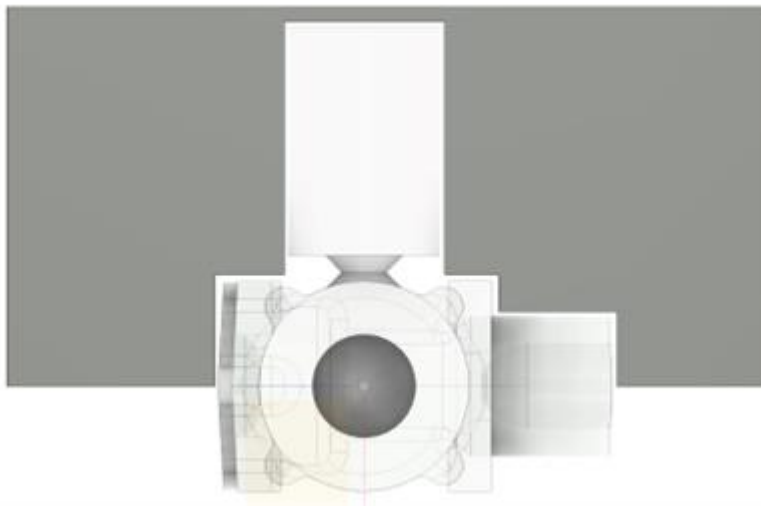


Figure 17

5.13.3. Cope weight (hold-down)

The cope must be heavier than the lifting force. Its sand volume is the cope box volume less the cavities it contains, here one casting and one riser (a single casting is poured in the mould, gated through branch C):

Quantity	Value
Cope box volume ($315 \times 400 \times 135$ mm)	17.01 dm ³
– Riser volume	0.189 dm ³
– Casting volume	0.339 dm ³
Net cope sand volume VUF	16.48 dm ³
Sand specific weight γ_{sand}	2.7 kg/dm ³
Cope weight $F_S = \gamma_{\text{sand}} \cdot VUF$	44.5 kg

The cope box height used here is 135 mm, the upper portion of the 250 mm total flask ($315 \times 400 \times 250$, standard UNI 6765-70). Only the cope sand bears down on the cavity, so the upper portion is the relevant volume. The sprue and runner cavities and the central core are neglected — a minor, slightly conservative simplification.

5.13.4. Verification

Lifting force F_{TOT}	Cope weight F_S	Result
22.1 kg	44.5 kg	$\checkmark F_S > F_{\text{TOT}} (\approx 2\times \text{margin})$

The weight of the cope exceeds the metallostatic lifting force by roughly a factor of two. The forces are therefore counteracted by the flask itself: no additional cope weights are required and the clamping screws are not load-critical for lifting. Because the rectangular approximation tends to under-state the contribution of surfaces at large radius, the true margin is the binding figure; the comfortable factor of two leaves the conclusion safe.

5.13.5. Summary

The metallostatic lifting force on the cope is $F_{\text{TOT}} = 22.1$ kg, evaluated as the weight of the metal column above the casting's upward surfaces using the rectangular (extruded-profile) approximation. The cope sand weight is $F_S = 44.5$ kg. Since $F_S > F_{\text{TOT}}$ with about a $2\times$ margin, the upper flask holds the mould closed without added weights or load-bearing clamps.

6. Conclusion

This project developed the complete sand-casting process design for the angled three-port junction housing, a medium-strength cast-steel component (grade FeG 520, ≈ 520 MPa). Starting from the fully-machined CAD geometry, every technological element required to produce the part by sand casting was defined in sequence — casting-blank geometry, feeding (risers), gating, molding and mould stability — so that the as-cast part can be finished to drawing by the subsequent machining cycle.

In the casting-blank design, the small holes and the fine internal machined features were left full (drilled and reamed later) rather than cast, since their tight tolerances and surface finish cannot be obtained as-cast, and the slender central passages were filled solid after the wall-thickness/central-core feasibility check. A uniform 4 mm machining allowance (80–180 mm size class) was applied to the surfaces to be finished, the Heuver inscribed-circle method was used to keep wall-thickness variation within the ~ 30 % limit, constant $R = 0.5$ mm fillets were added to all internal junctions, and a single horizontal parting plane with a 3° draft was adopted. As a result the blank envelope grew from the 129 mm finished length to 140.25 mm.

Feeding was designed from a four-zone thermal-modulus analysis ($M_A = 5.4$, $M_B = 8.5$, $M_C = 8.5$, $M_D = 6.8$ mm), which located the riser on branch C — the natural feed path into the heavy central hub. A single cylindrical riser with an insulating/exothermic sleeve ($M_M = 1.0$ cm, $\varnothing 54 \times 81$ mm) was selected by feeding capacity from ASSO FOND Table 6 rather than by modulus alone; this cut the riser metal by roughly 43 % relative to the modulus-only pick while still covering the casting volume, markedly improving yield. Chills on the hub and extremities, the riser sleeve and generous junction fillets were specified to correct the two non-ideal solidification steps ($A \rightarrow B$ at +58 % and the balanced $B \rightarrow C$ at +1 %) and to anchor a directional front toward the riser.

The metal was delivered through a single-ingate pressurised gating system (ratio $S_S : S_R : S_G = 4 : 3 : 2$) on branch C, sized for a poured weight of 4.117 kg and a 4.06 s pouring time, giving a gate velocity of 1.005 m/s and a required choke area of 129.4 mm². The standard sections finally chosen — sprue $\varnothing 20$ mm, runner 20×24 mm and ingate 18×18 mm — all sit at or above their calculated minima, keeping the ingate as the choke and the system back-filled. The component is moulded by match-plate (single cavity) in synthetic resin-bonded sand for the 500-piece batch, in a standard $315 \times 400 \times 250$ mm flask (UNI 6765-70); the metallostatic lifting force on the cope (≈ 22.1 kgf) is exceeded by the cope weight by about a factor of two, so the flask itself holds the mould down and no additional clamps or weights are required.

Overall, the design delivers a manufacturable, internally sound casting: directional solidification toward a correctly placed and well-sized riser, controlled and calm mould filling, and adequate mould stability — while keeping a good casting yield. The as-cast blank carries the appropriate allowances so that, after the planned machining operations, the finished junction housing meets its dimensional and surface-finish requirements.